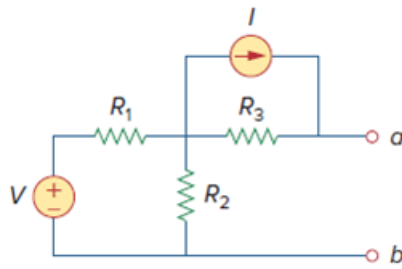


Problem

Using Fig. 4.102, design a problem that will help other students better understand Thevenin equivalent circuits.

Figure. 4.102:



Step-by-step solution

Step 1 of 6

Refer to the Figure 4.102 in the text book.

Assume the elements values in the circuit as follows.

$$V = 40 \text{ V}$$

$$I = 3 \text{ A}$$

$$R_1 = 6 \, \Omega$$

$$R_2 = 4 \, \Omega$$

$$R_3 = 6 \, \Omega$$

Draw the following circuit diagram.

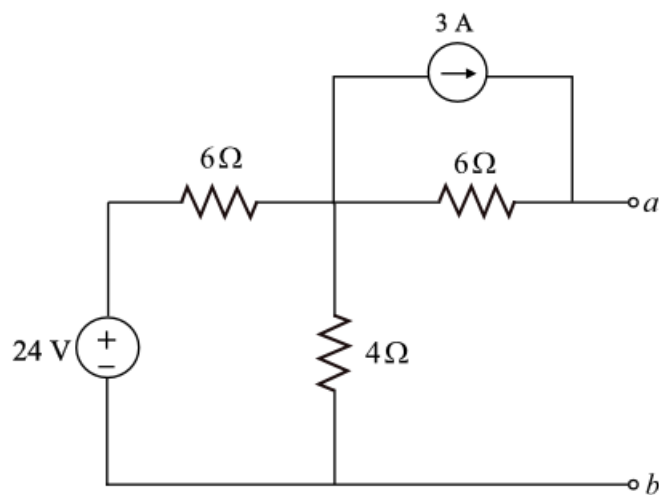


Figure 1

Step 2 of 6

Draw the following circuit diagram to determine the Thevenin voltage.

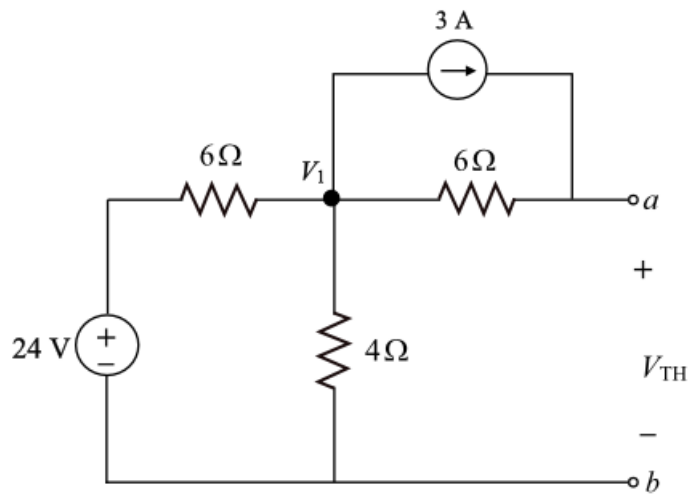


Figure 2

Step 3 of 6

Use nodal analysis in Figure 2.

Apply Kirchhoff's current law at node V_1 .

$$\frac{V_1 - 24}{6} + \frac{V_1}{4} + 3 + \frac{V_1 - V_{TH}}{6} = 0$$

$$V_1 \left[\frac{1}{6} + \frac{1}{4} + \frac{1}{6} \right] - \frac{V_{TH}}{6} = -3$$

$$V_1 \left[\frac{14}{24} \right] - \frac{V_{TH}}{6} = -3$$

$$14V_1 - 4V_{TH} = -72 \quad \dots\dots (1)$$

Apply Kirchhoff's current law at node V_{TH} .

$$\frac{V_{TH} - V_1}{6} - 3 = 0$$

$$V_{TH} - V_1 = 18 \quad \dots\dots (2)$$

Solve equation (1) and equation (2).

$$V_{TH} = 18 \text{ V}$$

Therefore, the value of Thevenin voltage is $V_{TH} = 18 \text{ V}$.

Step 4 of 6

Replace voltage source with short circuit and current source with open circuit to find the Thevenin resistance.

Draw the following circuit diagram to determine the Thevenin resistance.

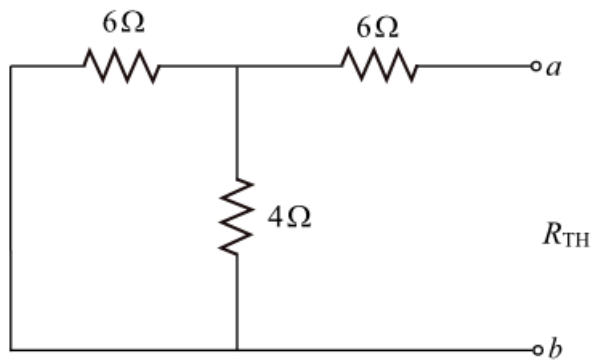


Figure 3

Step 5 of 6

Determine the Thevenin resistance using Figure 3.

$$\begin{aligned} R_{TH} &= (6 \parallel 4) + 6 \\ &= \left(\frac{24}{6+4} \right) + 6 \\ &= 2.4 + 6 \\ &= 8.4 \Omega \end{aligned}$$

Therefore, the value of Thevenin resistance is $R_{TH} = 8.4 \Omega$.

Step 6 of 6

Draw the Thevenin equivalent circuit as shown in Figure 4.

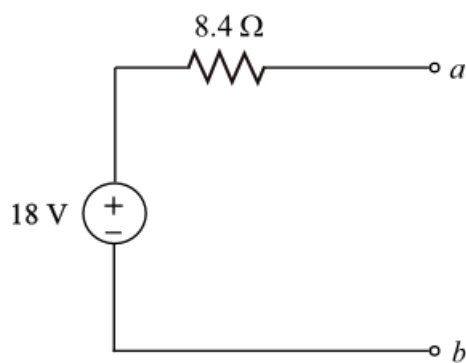


Figure 4